

1. For $f(x) = 2^{-x}k$; $x = 1, 2, 3, \dots$, determine the constant k for which $f(x)$ is the *pmf* for a random variable X .

$$\sum_{x=1}^{\infty} 2^{-x}k = 1 \rightarrow k \left(\frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \dots \right) = 1 \rightarrow k \frac{\frac{1}{2}}{1 - \frac{1}{2}} = 1 \rightarrow k = 1$$

$$f(x) = 2^{-x}; x = 1, 2, 3, \dots$$

2. For $f(x) = \frac{cx^2}{1435}$; $x = 1, 2, \dots, 20$, determine the constant c so that $f(x)$ satisfies the conditions of being *pmf* for a random variable X .

$$\sum_{x=1}^{20} \frac{cx^2}{1435} = 1 \rightarrow \frac{c}{1435} (1^2 + 2^2 + \dots + 20^2) = 1 \rightarrow \frac{c}{1435} \frac{20(20+1)(2 \times 20 + 1)}{6} = 1 \rightarrow c = \frac{1435 \times 6}{20 \times 21 \times 41} \rightarrow c = \frac{1}{2}$$

$$f(x) = \frac{x^2}{2870}; x = 1, 2, \dots, 20$$

3. The probability distribution table for a random variable X is shown below. Given that $E(X) = 0.28$, find the value m and n .

x	-2	-1	0.5	1	2
$P(X = x)$	0.12	m	n	0.16	0.3

$$\sum P(x) = 1 \rightarrow 0.12 + m + n + 0.16 + 0.3 = 1 \rightarrow m + n = 0.42$$

$$E(X) = 0.28 \rightarrow -0.24 - m + 0.5n + 0.16 + 0.6 = 0.28 \rightarrow -m + 0.5n = -0.24$$

$$\text{Solving system of linear equations, } m = 0.3, n = 0.12$$

4. Let the random variable X have the *pmf* $f(x) = \frac{(|x|-1)^2}{9}$; $x = -3, 0, 3$. Compute the mean, standard deviation, and $V(3X + 1)$.

x	-3	0	3
$P(X = x)$	$\frac{4}{9}$	$\frac{1}{9}$	$\frac{4}{9}$

$$E[X] = -3 \times \frac{4}{9} + 0 \times \frac{1}{9} + 3 \times \frac{4}{9} = 0$$

$$E[X^2] = (-3)^2 \times \frac{4}{9} + 0^2 \times \frac{1}{9} + 3^2 \times \frac{4}{9} = 8$$

$$V(X) = E(X^2) - \{E(X)\}^2 = 8 - 0^2 = 8$$

$$\mu = E[X] = 0$$

$$\sigma = \sqrt{V(X)} = \sqrt{8}$$

$$V(3X + 1) = 9V(X) = 9 \times 8 = 72$$

5. A fair 4 sided die, numbered 2, 3, 4, and 5 is rolled twice. The random variable X is the sum of the two outcomes on which the die comes to rest. (i) Show that $P(6 \leq X \leq 7) = \frac{7}{16}$, (ii) Draw up the probability distribution table for X , and find $P(X > 6)$.

$$D_1 = \{2,3,4,5\} \text{ and } D_2 = \{2,3,4,5\}$$

$$S = \left\{ \begin{pmatrix} (2,2) & (2,3) & (2,4) & (2,5) \\ (3,2) & (3,3) & (3,4) & (3,5) \\ (4,2) & (4,3) & (4,4) & (4,5) \\ (5,2) & (5,3) & (5,4) & (5,5) \end{pmatrix} \right\} \rightarrow n(S) = 16$$

X	4	5	6	7	8	9	10
$P(X)$	$\frac{1}{16}$	$\frac{2}{16}$	$\frac{3}{16}$	$\frac{4}{16}$	$\frac{3}{16}$	$\frac{2}{16}$	$\frac{1}{16}$

$$(i) \quad P(6 \leq X \leq 7) = P(X = 6) + P(X = 7) = \frac{3}{16} + \frac{4}{16} = \frac{7}{16}$$

$$(ii) \quad P(X > 6) = P(X = 7) + P(X = 8) + P(X = 9) + P(X = 10) = \frac{4}{16} + \frac{3}{16} + \frac{2}{16} + \frac{1}{16} = \frac{10}{16}$$

6. Consider $f(x) = P(X = x) = \begin{cases} 0.3; x = 1 \\ 0.5; x = 2 \\ 0.2; x = 3 \end{cases}$ as the *pmf* of the random variable X to compute $E[X(3 - X) + 2]$ and $V(2 - 3X)$.

$$E(X) = 1 \times 0.3 + 2 \times 0.5 + 3 \times 0.2 = 1.9$$

$$E(X^2) = 1^2 \times 0.3 + 2^2 \times 0.5 + 3^2 \times 0.2 = 4.1$$

$$E[X(3 - X) + 2] = 3E(X) - E(X^2) + 2 = 3 \times 1.9 - 4.1 + 2 = 3.6$$

$$V(2 - 3X) = 9V(X) = 9[E(X^2) - \{E(X)\}^2] = 9\{4.1 - 1.9^2\} = 4.41$$

7. In a bet, the betting person wins \$1, \$2 and \$3 with probabilities 0.25, 0.2 and 0.15, and loses \$1 with probability 0.4 for each \$1 bet. Find μ , σ , and $E[10 - X^2]$.

x	1	2	3	-1
$P(X = x)$	0.25	0.2	0.15	0.4

$$E[X] = 1 \times 0.25 + 2 \times 0.2 + 3 \times 0.15 + (-1) \times 0.4 = 0.7$$

$$E[X^2] = 1^2 \times 0.25 + 2^2 \times 0.2 + 3^2 \times 0.15 + (-1)^2 \times 0.4 = 2.8$$

$$\mu = E[X] = 0.7$$

$$\sigma = \sqrt{E(X^2) - \{E(X)\}^2} = \sqrt{2.8 - (0.7)^2} = 1.52$$

$$E[10 - X^2] = 10 - E[X^2] = 10 - 2.8 = 7.2$$

8. Let a random experiment be the casting of a pair of fair *four-sided* dice and let X equal the sum of two outcomes. With reasonable assumptions, find *pmf* of X .

$$S = \left\{ \begin{pmatrix} (1,1) & (1,2) & (1,3) & (1,4) \\ (2,1) & (2,2) & (2,3) & (2,4) \\ (3,1) & (3,2) & (3,3) & (3,4) \\ (4,1) & (4,2) & (4,3) & (4,4) \end{pmatrix} \right\} \rightarrow n(S) = 16$$

X	2	3	4	5	6	7	8
$P(X)$	$\frac{1}{16}$	$\frac{2}{16}$	$\frac{3}{16}$	$\frac{4}{16}$	$\frac{3}{16}$	$\frac{2}{16}$	$\frac{1}{16}$

$$f(x) = \frac{y_{ext} - |x - x_{ext}|}{m} = \frac{5 - |x - 4|}{16}; x = 2, 3, \dots, 8$$

9. Let a random experiment be the casting of a pair of fair *five-sided* dice and let X equal the maximum of two outcomes. With reasonable assumptions, find *pmf* of X . Also, find the standard deviation of X .

X	1	2	3	4	5
$P(X)$	$\frac{1}{25}$	$\frac{3}{25}$	$\frac{5}{25}$	$\frac{7}{25}$	$\frac{9}{25}$

$$\frac{f(x) - \frac{1}{25}}{\frac{9}{25} - \frac{1}{25}} = \frac{x-1}{5-1} \rightarrow \frac{25f(x)-1}{8} = \frac{x-1}{4} \rightarrow 25f(x) - 1 = 2x - 2 \rightarrow f(x) = \frac{2x-1}{25}; x = 1, 2, 3, 4, 5$$

$$E[X] = 1 \times \frac{1}{25} + 2 \times \frac{3}{25} + 3 \times \frac{5}{25} + 4 \times \frac{7}{25} + 5 \times \frac{9}{25} = \frac{95}{25} = 3.8$$

$$E[X^2] = 1^2 \times \frac{1}{25} + 2^2 \times \frac{3}{25} + 3^2 \times \frac{5}{25} + 4^2 \times \frac{7}{25} + 5^2 \times \frac{9}{25} = \frac{395}{25} = 15.8$$

$$\sigma = \sqrt{E(X^2) - \{E(X)\}^2} = \sqrt{15.8 - (3.8)^2} = 1.166$$

10. Let a random experiment be the casting of a pair of fair *six-sided* dice and let X equal the minimum of two outcomes. With reasonable assumptions, find *pmf* of X .

X	1	2	3	4	5	6
$P(X)$	$\frac{11}{36}$	$\frac{9}{36}$	$\frac{7}{36}$	$\frac{5}{36}$	$\frac{3}{36}$	$\frac{1}{36}$

$$\frac{f(x) - \frac{3}{36}}{\frac{9}{36} - \frac{3}{36}} = \frac{x-5}{2-5} \rightarrow \frac{36f(x)-1}{6} = \frac{x-3}{-3} \rightarrow 36f(x) - 1 = -2x + 12 \rightarrow f(x) = \frac{13-2x}{36}; x = 1, 2, 3, 4, 5, 6$$

11. A delivery company gives drivers a daily performance bonus based on the number of successful deliveries made in a day. The number of deliveries completed varies due to traffic, weather, and demand. The company's data show the following probability distribution for the number of deliveries a driver completes in a day, and the corresponding bonus earned.

Number of deliveries	5	6	7	8	9
Probability	0.1	0.2	0.4	0.2	0.1
Bonus earned (in Tk)	200	250	300	350	400

- (i) Find the average number of deliveries a driver completes per day.
(ii) Find the standard deviation of the daily bonus (in Tk.) a driver earns.

$$E[\text{Number}] = 5 \times 0.1 + 6 \times 0.2 + 7 \times 0.4 + 8 \times 0.2 + 9 \times 0.1 = 7$$

$$E[\text{Bonus}] = 200 \times 0.1 + 250 \times 0.2 + 300 \times 0.4 + 350 \times 0.2 + 400 \times 0.1 = 300$$

$$E[\text{Bonus}^2] = 200^2 \times 0.1 + 250^2 \times 0.2 + 300^2 \times 0.4 + 350^2 \times 0.2 + 400^2 \times 0.1 = 93000$$

$$\mu_{\text{Number}} = E[\text{Number}] = 7$$

$$\sigma_{\text{Bonus}} = \sqrt{E[\text{Bonus}^2] - \{E[\text{Bonus}]\}^2} = \sqrt{93000 - 300^2} = 54.77$$

12. Let a random variable X be the number of days that an employee needs to be in leave satisfying the probability function $f(x) = \frac{1}{15}(6 - x)$; $x = 1, 2, 3, 4, 5$. If the employee is to receive \$200 for the first two day, \$100 for the next two day, and have to return \$25 for the final day, what is the average payment for the hospitalization of the employees? What is the standard deviation of that payment?

x	1	2	3	4	5
$P(X = x)$	$\frac{5}{15}$	$\frac{4}{15}$	$\frac{3}{15}$	$\frac{2}{15}$	$\frac{1}{15}$
Payment	\$200	\$400	\$500	\$600	\$575

$$E(X) = 200 \times \frac{5}{15} + 400 \times \frac{4}{15} + 500 \times \frac{3}{15} + 600 \times \frac{2}{15} + 575 \times \frac{1}{15} = \frac{5875}{15}$$

$$E(X^2) = 200^2 \times \frac{5}{15} + 400^2 \times \frac{4}{15} + 500^2 \times \frac{3}{15} + 600^2 \times \frac{2}{15} + 575^2 \times \frac{1}{15} = \frac{2640625}{15}$$

$$\mu = E(X) = \frac{5875}{15} = 391.67$$

$$\sigma = \sqrt{E(X^2) - \{E(X)\}^2} = \sqrt{\frac{2640625}{15} - \left\{\frac{5875}{15}\right\}^2} = 150.46$$

13. Suppose there are 5 defective items in a lot of 50 items. A sample of size 10 is taken at random without replacement. Let X denote the number of defective items in the sample. Find the probability that the sample contains (i) at most one defective item (ii) at least four defective items (iii) at least an item is defective.

$$N = 50, N_1 = 5, N_2 = 45, n = 10$$

$$f(x) = \frac{{}^5_x ({}^{45}_{10-x})}{{}^{50}_{10}}; x = 0,1,2,3,4,5$$

$$P(X \leq 1) = P(X = 0) + P(X = 1) = \frac{{}^5_0 ({}^{45}_{10})}{{}^{50}_{10}} + \frac{{}^5_1 ({}^{45}_9)}{{}^{50}_{10}} = 0.3106 + 0.4313 = 0.7419$$

$$P(X \geq 4) = P(X = 4) + P(X = 5) = \frac{{}^5_4 ({}^{45}_6)}{{}^{50}_{10}} + \frac{{}^5_5 ({}^{45}_5)}{{}^{50}_{10}} = 0.00396 + 0.00012 = 0.00408$$

$$P(X \geq 1) = 1 - P(X = 0) = 1 - \frac{{}^5_0 ({}^{45}_{10})}{{}^{50}_{10}} = 1 - 0.3106 = 0.6894$$

14. A bag contains 7 balls with 3 blue balls, and the rest of them are of other colors. 4 balls are selected at random from the bag, without replacement. Let X denote the number of blue balls selected. Now, (i) construct a table to show the probability distribution of X (ii) find the mean and variance of X .

$$N = 7, N_1 = 3, N_2 = 4, n = 4$$

$$f(x) = \frac{{}^3_x ({}^4_{4-x})}{{}^7_4}; x = 0,1,2,3$$

x	0	1	2	3
$P(X = x)$	0.1667	0.5000	0.3000	0.0333

$$E(X) = 0 \times 0.1667 + 1 \times 0.5000 + 2 \times 0.3000 + 3 \times 0.0333 = 1.1999$$

$$E(X^2) = 0^2 \times 0.1667 + 1^2 \times 0.5000 + 2^2 \times 0.3000 + 3^2 \times 0.0333 = 1.9997$$

$$\mu = E(X) = 1.1999$$

$$\sigma^2 = E(X^2) - \{E(X)\}^2 = 1.9997 - 1.1999^2 = 0.5599$$

15. During a software testing phase, each module has a 0.1 probability of containing a critical bug. If a tester independently selects 12 modules, find the probability that no modules have critical bugs. Also, find the probability that between 2 and 4 modules have the critical bug.

$$n = 12, p = 0.1, q = 0.9$$

$$f(x) = {}^{12}C_x (0.1)^x (0.9)^{12-x}; x = 0,1, \dots, 12$$

$$P(X = 0) = {}^{12}C_0 (0.1)^0 (0.9)^{12} = 0.2824$$

$$P(2 \leq X \leq 4) = P(X = 2) + P(X = 3) + P(X = 4)$$

$$= {}^{12}C_2 (0.1)^2 (0.9)^{10} + {}^{12}C_3 (0.1)^3 (0.9)^9 + {}^{12}C_4 (0.1)^4 (0.9)^8 = 0.2301 + 0.0852 + 0.0213 = 0.3366$$

16. It is claimed that 5% of the birds in a particular region have an infectious disease. Suppose that 10 birds are selected at random. Let X be the number of birds that have the disease. Assuming independence, how X is distributed? Find that (i) exactly 3 (ii) at most 8 birds are infected.

$$n = 10, p = 0.05, q = 0.95$$

$$f(x) = {}^{10}C_x (0.05)^x (0.95)^{10-x}; x = 0,1,2,3, \dots, 10$$

$$P(X = 3) = {}^{10}C_3 (0.05)^3 (0.95)^7 = 0.0105$$

$$P(X \leq 8) = 1 - [P(X = 9) + P(X = 10)]$$

$$= 1 - [{}^{10}C_9 (0.05)^9 (0.95)^1 + {}^{10}C_{10} (0.05)^{10} (0.95)^0] = 1 - 1.8652 \times 10^{-11}$$

17. A company produces bags of sugar. An inspector finds that, on average 10% of the bags are underweight. 9 bags are chosen at random, find the probability that (i) fewer than 3 (ii) more than 1 of these bags are underweight. Also, find the mean and standard deviation.

$$n = 9, \quad p = 0.1, \quad q = 0.9$$

$$f(x) = {}^9C_x (0.1)^x (0.9)^{9-x}; \quad x = 0, 1, 2, \dots, 9$$

$$P(X < 3) = P(X = 0) + P(X = 1) + P(X = 2)$$

$$= {}^9C_0 (0.1)^0 (0.9)^9 + {}^9C_1 (0.1)^1 (0.9)^8 + {}^9C_2 (0.1)^2 (0.9)^7 = 0.3874 + 0.3874 + 0.1722 = 0.947$$

$$P(X > 1) = 1 - [P(X = 0) + P(X = 1)] = 1 - [{}^9C_0 (0.1)^0 (0.9)^9 + {}^9C_1 (0.1)^1 (0.9)^8]$$

$$= 1 - (0.3874 + 0.3874) = 0.2254$$

$$\mu = np = 9 \times 0.1 = 0.9$$

$$\sigma = \sqrt{npq} = \sqrt{9 \times 0.1 \times 0.9} = 0.9$$

18. A garden shop sells polyanthus plants in boxes, each box containing the same number of plants. The number of plants per box that produce yellow flowers has a binomial distribution with a mean of 11 and a variance of 4.95. Then, (i) find the number of plants per box (ii) find the probability that a box contains exactly 12 plants which produce yellow flowers.

$$\text{Mean, } \mu = np = 11 \quad \dots\dots\dots (i)$$

$$\text{Variance, } \sigma^2 = npq = 4.95 \quad \dots\dots\dots (ii)$$

$$(ii) \div (i) \Rightarrow q = 0.45 \text{ and hence } p = 0.55,$$

$$\text{So, } n = \frac{11}{p} = \frac{11}{0.55} = 20 \text{ [from (i)]}$$

$$f(x) = {}^{20}C_x (0.55)^x (0.45)^{20-x}; \quad x = 0, 1, 2, \dots, 20$$

$$P(X = 12) = {}^{20}C_{12} (0.55)^{12} (0.45)^8 = 0.1620$$

19. A network server receives an average of 4 data requests per second. Assuming requests arrive randomly under a Poisson process, find the probability that in two seconds the server receives at least 1 request. Also, find the probability that the number of requests arrive in the next two seconds is 5 or 6.

$$\lambda = 4 \times 2 = 8$$

$$f(x) = \frac{e^{-8} 8^x}{x!}; \quad x = 0, 1, 2, 3, \dots$$

$$P(X \geq 1) = 1 - P(X = 0) = 1 - \frac{e^{-8} 8^0}{0!} = 1 - e^{-8} = 0.9997$$

$$P(X = 5) + P(X = 6) = \frac{e^{-8} 8^5}{5!} + \frac{e^{-8} 8^6}{6!} = 0.0916 + 0.1221 = 0.2137$$

20. A drop of water falls randomly from a leaking tap at a constant average rate of 5.2 per minute. Find the probability that at least 3 drops fall during a randomly chosen 30-second period. Also, find the mean and standard deviation of the drops.

$$\lambda = 30 \times \frac{5.2}{60} = 2.6$$

$$f(x) = \frac{2.6^x e^{-2.6}}{x!}; x = 0, 1, 2, \dots$$

$$\begin{aligned} P(X \geq 3) &= 1 - P(X < 3) = 1 - [P(X = 0) + P(X = 1) + P(X = 2)] \\ &= 1 - \left[\frac{2.6^0 e^{-2.6}}{0!} + \frac{2.6^1 e^{-2.6}}{1!} + \frac{2.6^2 e^{-2.6}}{2!} \right] = 1 - (0.0743 + 0.1931 + 0.2510) = 0.4816 \end{aligned}$$

$$\mu = \lambda = 2.6$$

$$\sigma = \sqrt{\lambda} = \sqrt{2.6}$$

21. Computer breakdowns occur randomly on average once every 48 hours of use. (i) Calculate the probability that there will be fewer than 4 breakdowns in 60 hours of use, (ii) There will be no breakdowns in 24 hours of use.

$$\begin{aligned} \text{(i)} \quad \lambda &= \frac{60}{48} = 1.25 \rightarrow f(x) = \frac{e^{-1.25} 1.25^x}{x!}; x = 0, 1, 2, 3, \dots \\ P(X < 4) &= P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) \\ &= \frac{e^{-1.25} 1.25^0}{0!} + \frac{e^{-1.25} 1.25^1}{1!} + \frac{e^{-1.25} 1.25^2}{2!} + \frac{e^{-1.25} 1.25^3}{3!} = 0.9617 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \lambda &= \frac{24}{48} = 0.5 \rightarrow f(x) = \frac{e^{-0.5} 0.5^x}{x!}; x = 0, 1, 2, 3, \dots \\ P(X = 0) &= \frac{e^{-0.5} 0.5^0}{0!} = 0.6065 \end{aligned}$$

22. Between 7 PM and 11 PM, arrivals of patients at the casualty department of a hospital occur at random at an average rate of 6 per hour. (i) Find the probability that, during any period of one hour between 7 PM and 11 PM, exactly 5 people will arrive, (ii) A patient arrives at exactly 10.15 PM; find the probability that at least one more patient arrives before 10.35 PM.

$$\begin{aligned} \text{(i)} \quad \lambda &= 6 \rightarrow f(x) = \frac{e^{-6} 6^x}{x!}; x = 0, 1, 2, 3, \dots \\ P(X = 5) &= \frac{e^{-6} 6^5}{5!} = 0.1606 \\ \text{(ii)} \quad \lambda &= 6 \times \frac{20}{60} = 2 \rightarrow f(x) = \frac{e^{-2} 2^x}{x!}; x = 0, 1, 2, 3, \dots \\ P(X \geq 1) &= 1 - P(X = 0) = 1 - \frac{e^{-2} 2^0}{0!} = 0.8647 \end{aligned}$$

23. The time (in minutes) taken by a computer to complete a data backup operation is modeled as a continuous random variable X with the probability density function $f(x) = k(10 - x); 0 \leq x \leq 10$.

- Determine the value of k that makes $f(x)$ a valid probability density function.
- Find the cumulative distribution function $F(x)$.
- Find the probability that the backup takes less than 5 minutes.
- Compute the average time for the backup to complete.
- Determine the median backup time.

$$\int_0^{10} k(10-x)dx = 1 \rightarrow k \left[10x - \frac{x^2}{2} \right]_0^{10} = 1 \rightarrow k[100 - 50] = 1 \rightarrow 50k = 1 \rightarrow k = \frac{1}{50}$$

$$f(x) = \frac{1}{50}(10-x); 0 \leq x \leq 10$$

$$F(x) = P(X \leq x) = \int_0^x \frac{1}{50}(10-w)dw = \frac{1}{50} \left[10w - \frac{w^2}{2} \right]_0^x = \frac{1}{100}[20x - x^2]; 0 \leq x \leq 10$$

$$P(X < 5) = F(5) = \frac{1}{100}[100 - 25] = 0.75$$

$$\mu = E[X] = \int_0^{10} x \left\{ \frac{1}{50}(10-x) \right\} dx = \frac{1}{50} \int_0^{10} (10x - x^2) dx = \frac{1}{50} \left[5x^2 - \frac{x^3}{3} \right]_0^{10} = \frac{1}{50} \left[500 - \frac{1000}{3} \right] = 3.33$$

$$F(m) = 0.5 \rightarrow \frac{1}{100}[20m - m^2] = 0.5 \rightarrow 20m - m^2 = 50 \rightarrow m^2 - 20m + 50 = 0 \rightarrow m = 10 \pm 5\sqrt{2}$$

$$m = 2.93 \in 0 \leq x \leq 10$$

24. The life X (in years) of a voltage regulator of a car has the *pdf* $f(x) = 3x^2e^{-x^3}$ defined in $0 \leq x < \infty$. What is the probability that it will last at least 5 years? If it has lasted for 5 years, find the conditional probability that it will last at least 2 years more. Also, find the 3rd quartile of the of the life time for that voltage regulator.

$$\begin{aligned} F(x) = P(X \leq x) &= \int_0^x 3w^2 e^{-w^3} dw = \int_0^{w^3} e^{-z} dz & \left| \begin{array}{l} w^3 = z \Rightarrow 3w^2 dw = dz \\ w = 0, x \rightarrow z = 0, x^3 \end{array} \right. \\ &= [-e^{-z}]_0^{w^3} = 1 - e^{-x^3}; 0 < x < \infty \end{aligned}$$

$$P(X \geq 5) = 1 - P(X < 5) = 1 - F(5) = 1 - [1 - e^{-5^3}] = e^{-125}$$

$$P(X \geq 7) = 1 - P(X < 7) = 1 - F(7) = 1 - [1 - e^{-7^3}] = e^{-243}$$

$$P(X \geq 7 | X \geq 5) = \frac{P(X \geq 5 \cap X \geq 7)}{P(X \geq 5)} = \frac{P(X \geq 7)}{P(X \geq 5)} = \frac{e^{-243}}{e^{-125}} = e^{-118}$$

$$F(q) = 0.75 \rightarrow 1 - e^{-q^3} = 0.75 \rightarrow e^{-q^3} = \ln(0.25) \rightarrow q = -\sqrt[3]{\ln(0.25)} \rightarrow q = 1.115$$

25. Assume the *pdf* of X be $f(x) = ke^{-\frac{x}{2}}; 0 \leq x < \infty$. Then, find (i) the value of k (ii) 1st quartile of X (iii) $P(X \geq 2)$.

$$\int_0^{\infty} ke^{-\frac{x}{2}} dx = 1 \rightarrow \left[-2ke^{-\frac{x}{2}} \right]_0^{\infty} = 1 \rightarrow 2k = 1 \rightarrow k = \frac{1}{2}$$

$$f(x) = \frac{1}{2}e^{-\frac{x}{2}}; 0 \leq x < \infty$$

$$F(x) = P(X \leq x) = \int_0^x \frac{1}{2}e^{-\frac{w}{2}} dw = \left[-e^{-\frac{w}{2}} \right]_0^x = 1 - e^{-\frac{x}{2}}; 0 \leq x < \infty$$

$$F(q) = 0.25 \rightarrow 1 - e^{-\frac{q}{2}} = 0.25 \rightarrow e^{-\frac{q}{2}} = 0.75 \rightarrow q = -2 \ln(0.75) = 0.5754$$

$$P(X \geq 2) = 1 - P(X < 2) = 1 - F(2) = 1 - (1 - e^{-1}) = e^{-1}$$

26. For $f(x) = 2x^c$; $0 \leq x \leq 1$, find the constant c so that $f(x)$ is a *pdf* of a random variable X . Find the *cdf* of X and hence find $P(X > 0.75)$ and median of X .

$$\int_0^1 2x^c dx = 1 \rightarrow \left[\frac{2x^{c+1}}{c+1} \right]_0^1 = 1 \rightarrow \frac{2}{c+1} = 1 \rightarrow c+1 = 2 \rightarrow c = 1$$

$$f(x) = 2x; 0 \leq x \leq 1$$

$$F(x) = P(X \leq x) = \int_0^x 2w dw = [w^2]_0^x = x^2; 0 \leq x \leq 1$$

$$P(X > 0.75) = 1 - P(X \leq 0.75) = 1 - F(0.75) = 1 - (0.75)^2 = 0.4375$$

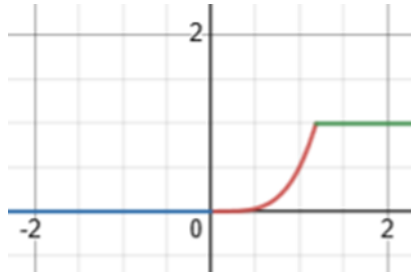
$$F(m) = 0.5 \rightarrow m^2 = 0.5 \rightarrow m = \sqrt{0.5} \rightarrow m = 0.7071$$

27. Assume the *pdf* of X be $f(x) = 2x^3$; $0 \leq x \leq c$, find the constant c . Find and sketch the *cdf* of X . Estimate the mean and standard deviation and the 3rd quartile of the distribution. Also, find $P(X \geq 2)$.

$$\int_0^c 2x^3 dx = 1 \rightarrow \left[\frac{x^4}{2} \right]_0^c = 1 \rightarrow c^4 = 2 \rightarrow c = \sqrt[4]{2}$$

$$f(x) = 2x^3; 0 \leq x \leq \sqrt[4]{2}$$

$$F(x) = P(X \leq x) = \int_0^x 2w^3 dw = \frac{w^4}{2} \Big|_0^x = \frac{x^4}{2}; 0 \leq x \leq \sqrt[4]{2}$$



$$E(X) = \int_0^{\sqrt[4]{2}} x (2x^3) dx = 2 \int_0^{\sqrt[4]{2}} x^4 dx = \frac{2x^5}{5} \Big|_0^{\sqrt[4]{2}} = \frac{4\sqrt[4]{2}}{5}$$

$$E(X^2) = \int_0^{\sqrt[4]{2}} x^2 (2x^3) dx = 2 \int_0^{\sqrt[4]{2}} x^5 dx = \frac{x^6}{3} \Big|_0^{\sqrt[4]{2}} = \frac{2\sqrt[4]{2}}{3}$$

$$\mu = E(X) = \frac{4\sqrt[4]{2}}{5}$$

$$\sigma = \sqrt{E(X^2) - \{E(X)\}^2} = \sqrt{\frac{2\sqrt[4]{2}}{3} - \left(\frac{4\sqrt[4]{2}}{5}\right)^2}$$

$$F(q) = 0.75 \rightarrow \frac{q^4}{2} = 0.75 \rightarrow q^4 = 1.5 \rightarrow q = \sqrt[4]{1.5}$$

$$P(X \geq 2) = 1 - P(X < 2) = 1 - F(2) = 1 - 1 = 0$$

28. For $f(x) = \frac{3}{8}x^2$; $0 \leq x \leq l$, find the constant l so that $f(x)$ is a pdf of a random variable X . Find and sketch the *cdf* of x . Find the standard deviation of the distribution of X . Also, find $P(0.75 < X \leq 1.25)$ and the median of the distribution of X .

$$\int_0^l \frac{3}{8}x^2 dx = 1 \rightarrow \left[\frac{1}{8}x^3 \right]_0^l = 1 \rightarrow \frac{l^3}{8} = 1 \rightarrow l^3 = 8 \rightarrow l = 2$$

$$f(x) = \frac{3}{8}x^2; 0 \leq x \leq 2$$

$$E(X) = \int_0^2 x \left(\frac{3x^2}{8} \right) dx = \left[\frac{3x^4}{32} \right]_0^2 = \frac{3}{2} = 1.5$$

$$E(X^2) = \int_0^2 x^2 \left(\frac{3x^2}{8} \right) dx = \left[\frac{3x^5}{40} \right]_0^2 = \frac{12}{5} = 2.4$$

$$\sigma = \sqrt{E(X^2) - \{E(X)\}^2} = \sqrt{2.4 - (1.5)^2} = 0.39$$

$$F(x) = P(X \leq x) = \int_0^x \frac{3}{8}x^2 dw = \frac{w^3}{8} \Big|_0^x = \frac{x^3}{8}; 0 \leq x \leq 2$$



$$P(0.75 < X \leq 1.25) = F(1.25) - F(0.75) = \frac{1.25^3}{8} - \frac{0.75^3}{8} = 0.1914$$

$$F(m) = 0.5 \rightarrow \frac{m^3}{8} = 0.5 \rightarrow m^3 = 4 \rightarrow m = \sqrt[3]{4} = 1.59$$

29. Identify the value of k for which $f(x) = k(2x - 3x^2)$; $0 \leq x \leq \frac{1}{2}$ could be a *pdf*. Also, find the mode of the distribution.

$$\int_0^{\frac{1}{2}} k(2x - 3x^2) dx = 1 \rightarrow k \left[x^2 - x^3 \right]_0^{\frac{1}{2}} = 1 \rightarrow k \left[\frac{1}{4} - \frac{1}{8} \right] = 1 \rightarrow \frac{k}{8} = 1 \rightarrow k = 8$$

$$f(x) = 8(2x - 3x^2); 0 \leq x \leq \frac{1}{2}$$

$$f'(x) = 8(2 - 6x)$$

$$f'(m_0) = 0 \rightarrow 8(2 - 6m_0) = 0 \rightarrow m_0 = \frac{1}{3}$$

30. If $X \sim N(\mu, \sigma^2)$, then calculate $P(\mu - 2\sigma < X < \mu + 2\sigma)$.

$$P(\mu - 2\sigma < X < \mu + 2\sigma) = P\left(\frac{(\mu - 2\sigma) - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{(\mu + 2\sigma) - \mu}{\sigma}\right) = P(-2 < Z < 2)$$

$$= 2\phi(2) - 1 = 2 \times 0.9772 - 1 = 0.9544$$

31. The processing time for a batch of transactions on a data server is normally distributed with a mean of 2.5 seconds and a standard deviation of 0.4 seconds. Find the probability that a transaction takes less than 2 seconds to process.

$$\mu = 2.5, \quad \sigma = 0.4$$

$$P(X < 2) = P\left(\frac{X-2.5}{0.4} < \frac{2-2.5}{0.4}\right) = P(Z < -1.25) = \phi(-1.25) = 1 - \phi(1.25) = 1 - 0.89435 = 0.10565$$

32. If the *pdf* of the normal variable X is $f(x) = \frac{1}{\sqrt{72\pi}} e^{-\frac{(x-25)^2}{72}}$, find a constant k such that $P(|Z| \leq k) = 0.9544$ and hence find the corresponding X .

$$f(x) = \frac{1}{\sqrt{72\pi}} e^{-\frac{(x-25)^2}{72}} = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$\mu = 25, \quad 2\sigma^2 = 72 \rightarrow \sigma^2 = 36 \rightarrow \sigma = 6$$

$$P(|Z| \leq k) = 0.9544 \rightarrow P(-k \leq Z \leq k) = 0.9544 \rightarrow 2\phi(k) - 1 = 0.9544 \rightarrow \phi(k) = 0.9772 \rightarrow z_k = 2$$

$$x_k = \mu + \sigma z_k = 25 + (6)(2) = 37$$

33. In a normal distribution with mean μ and standard deviation σ , if $P(X \geq 3.6) = 0.5$ and $P(X > 2.8) = 0.6554$, write down the value of μ and calculate the value of σ

$$P(X \geq 3.6) = 0.5 \rightarrow P\left(Z \geq \frac{3.6-\mu}{\sigma}\right) = 0.5 \rightarrow P\left(Z < \frac{3.6-\mu}{\sigma}\right) = 1 - 0.5 \rightarrow \phi\left(\frac{3.6-\mu}{\sigma}\right) = 0.5 \rightarrow \frac{3.6-\mu}{\sigma} = 0 \rightarrow \mu = 3.6$$

$$P(X > 2.8) = 0.6554 \rightarrow P\left(Z > \frac{2.8-3.6}{\sigma}\right) = 0.6554 \rightarrow P\left(Z > -\frac{0.8}{\sigma}\right) = 0.6554 \rightarrow P\left(Z < \frac{0.8}{\sigma}\right) = 0.6554$$

$$\rightarrow \phi\left(\frac{0.8}{\sigma}\right) = 0.6554 \rightarrow \frac{0.8}{\sigma} = 0.4 \rightarrow \sigma = 2$$

34. The distance in meters that a ball can be thrown by pupils at a particular school follows a normal distribution with mean 35 meter and standard deviation 11.6 meter. (i) Find the probability that a randomly chosen pupil can be throw a ball between 30 meter and 40 meter. (ii) The school gives a certificate to the 10% of pupils who throw further than a certain distance, find the least distance that must be thrown to qualify for a certificate.

If X is normally distributed with mean $\mu = 35$ and standard deviation $\sigma = 11.6$.

$$\begin{aligned} \text{(i)} \quad P(30 \leq X \leq 40) &= P\left(\frac{30-35}{11.6} \leq Z \leq \frac{40-35}{11.6}\right) = P(-0.43 \leq Z \leq 0.43) \\ &= 2\phi(0.43) - 1 = 2 \times 0.6664 - 1 = 0.3328 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad P(X > x) &= 0.1 \rightarrow P(X \leq x) = 0.9 \rightarrow P\left(Z \leq \frac{x-35}{11.6}\right) = 0.9 \rightarrow \phi\left(\frac{x-35}{11.6}\right) = 0.9 \\ &\rightarrow \frac{x-35}{11.6} = 1.28 \rightarrow x = 35 + 11.6 \times 1.28 = 49.848 \end{aligned}$$

35. The waiting time in a doctor's surgery is normally distributed with mean 15 minutes and standard deviation 4.2 minutes. (i) Find the probability that a patient has to wait less than 10 minutes to see the doctor, (ii) 10% of people wait longer than T minutes, find T , (iii) In a given that week, 200 people attend the surgery, estimate the number of these who wait more than 20 minutes.

If X is normally distributed with mean $\mu = 15$ and standard deviation $\sigma = 4.2$.

$$(i) \quad P(X < 10) = P\left(Z < \frac{10-15}{4.2}\right) = P(Z < -1.19) = 1 - \phi(1.19) = 1 - 0.88298 = 0.11702$$

$$(ii) \quad P(X > T) = 0.1 \rightarrow P(X \leq T) = 0.9 \rightarrow P\left(Z \leq \frac{T-15}{4.2}\right) = 0.9 \Rightarrow \phi\left(\frac{T-15}{4.2}\right) = 0.9 \rightarrow \frac{T-15}{4.2} = 1.28 \\ \rightarrow T = 15 + 4.2 \times 1.28 = 20.376$$

$$(iii) \quad P(X > 20) = P\left(Z > \frac{20-15}{4.2}\right) = P(Z > 1.19) = 1 - \phi(1.19) = 1 - 0.88298 = 0.11702$$

$$\text{Estimated number for 200 patients } E(X) = n \times p = 200 \times 0.11702 = 23.404 \approx 24$$

36. In a certain country the time taken for a common infection to clear up is normally distributed with mean μ days and standard deviation 2.6 days. 25% of these infections clear up in less than 7 days. (i) Find the value of μ , (ii) In another country the standard deviation of the time taken for the infection clear up remains the same, but the mean is 6.5 days, find the probability that in a randomly chosen case from this country, the infection takes longer than 6.2 days to clear up.

If X is normally distributed with unknown mean μ and standard deviation $\sigma = 2.6$.

$$(i) \quad P(X < 7) = 0.25 \rightarrow P\left(Z < \frac{7-\mu}{2.6}\right) = 0.25 \rightarrow P\left(Z < -\frac{7-\mu}{2.6}\right) = 1 - 0.25 \rightarrow P\left(Z < -\frac{7-\mu}{2.6}\right) = 0.75 \\ \rightarrow \phi\left(-\frac{7-\mu}{2.6}\right) = 0.75 \rightarrow -\frac{7-\mu}{2.6} = 0.67 \rightarrow \mu = 7 + 2.6 \times 0.67 = 8.742$$

$$(ii) \quad \text{For } \mu = 6.5, P(X > 6.2) = P\left(Z > \frac{6.2-6.5}{2.6}\right) = P(Z > -0.12) = P(Z < 0.12) = \phi(0.12) = 0.54776$$